DESCRIPTION OF COURSEWORK

Course Code	EEE301
Course Name	Signals and Systems
Lecturer	Siti Balqis Samdin
Academic Session	2023/04
Assessment Title	Amplitude modulation in a broadcast transmission system

A. Introduction/ Situation/ Background Information

Selective filters are important components in signal processing. A common requirement in signal processing is to modify the frequency contents of a continuous-time (CT) signal in a predefined manner. Surrounding noises and interferences during signal transmission could corrupt the information carried. A linear time-invariant system (LTI) system can compensate for the distortion introduced during the transmission.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 3 Explain the design of different filtering, sampling, and modulation processes and their practical applications. (Map-to-PLO2)

CLO 4 Apply modern simulation tools in analyzing and designing practical circuits. (Map-to-PLO5)

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.

iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All the assessment submitted must be the outcome of the student. Any form of academic misconduct is a serious offense which will be penalized by being given a zero mark for the entire assessment in question or part of the assessment in question. If there is more than one guilty party as in the case of collusion, both you and your collusion partner(s) will be subjected to the same penalty.

D. Instruction to Students

1. This assignment is a group assignment. Students are required to form a group of TWO (2) to THREE (3) to complete the assignment. Choose your group member.
2. The group must submit a technical report (typewritten) of no more than 15 pages (excluding appendices) with 1.5 line spacing. The report should include the system background/introduction, the methodology (includes concept diagram and components with justification), and the expected outcomes.
3. Include the assignment cover page with 'own work declaration' statements, rubrics, and Turnitin report (in appendix).
4. A group leader is required to submit the complete assignment onto MOODLE site. There will be two files of submission in MOODLE; i) Assignment report (.pdf), and ii) MATLAB/OCTAVE codes (.m).

   Put all the files in a .zip folder. Name the folder as ID#1_ID#2_ID#3-AssignmentEEE301.
5. The due date is on 6 July 2023, 11:59 pm.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	System design	30
2.	Report	30
3.	Simulation codes/algorithms	30
4.	Turnitin report	10
	TOTAL	**100**

F. Task(s)

Title: Amplitude Modulations in a broadcast transmission system.

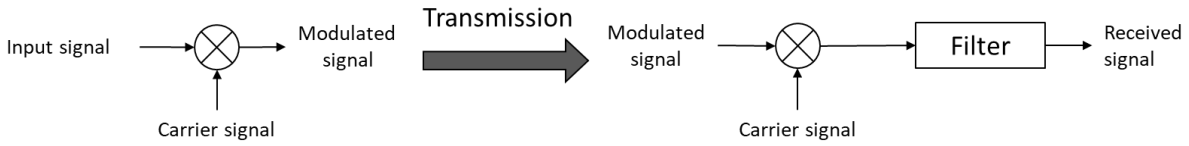


Figure 1: General block diagram of amplitude modulation system.

One of the applications of amplitude modulation can be found in a broadcast transmission system. The system develops to transmit the information from one end to another. Amplitude modulation systems have advantages, such as being easy to implement, cheaper, travelling long distances, and covering wider areas. Figure 1 shows the general block diagram of amplitude modulation that consist of the input signal, carrier signal, modulation, and filtering component.

Objective:

The objective is to develop a simulation system in MATLAB/OCTAVE that can transmit a clear message/signal from one end to another.

APPENDIX 1

MARKING RUBRICS

Component Title	Overall assignment component				Percentage (%)	15	
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)		
System Design (CLO3)	Comprehensive design with critical reasoning of the designed system/ component. Includes comprehensive theoretical explanation.	Adequate design with satisfactory reasoning of the designed system/ component. Includes good theoretical explanation.	Acceptable design with some minor misconceptions of the designed system/ component. Includes some theoretical explanation.	Show a limited design with weak reasoning of the designed system/ component. No theoretical explanation.	Poor design with no explanation.	30	
Report (CLO3)	An excellent report is comprehensive, well organized, well written, with clear explanation, has no errors, and is aided with references.	The report presented adequately organized and written, satisfactory explanations with minor errors and aided with references.	The report presented somewhat organized explanations written with some errors and a lack of references.	Presenting an unorganized report, and the written explanations have errors and no references.	The report is presented missing parts and has many errors.	30	
Codes/ Scripts (CLO4)	Excellently used simulation software to design/simulate the system. The code/ algorithm is well-prepared and written. The codes are aided with comments/explanations.	Adequately used simulation software to design/simulate part of the system. The code/ algorithm is satisfactorily prepared and written.	Somewhat appropriately use simulation software to design/simulate part of the system. The code/ algorithm is prepared and written with a minor error.	Use simulation software to design/ simulate the system. However, the codes/algorithms do not work.	Not using any simulation software.	30	
Turnitin (CLO4)	Similarity ≤ 15%	16 ≤Similarity ≤ 20%	21 ≤Similarity≤ 30%	31 ≤Similarity≤ 40%	Similarity≥ 41%	10	
					TOTAL	**100**	

Note to students: Please print out and attach this appendix together with the submission of coursework